

20260930 Quiz

for (MATH200) Differential Equations

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Problem 2.7.3. Consider the initial value problem

$$y' = 1 - t + 2y, \quad y(0) = 1.$$

- (a) Find approximate values of the solution at $t = 0.1, 0.2, 0.3, 0.4$ using Euler's method with $h = 0.1$.
- (b) Repeat part (a) with $h = 0.05$. Compare the results with those found in part (a).
- (c) Repeat part (a) with $h = 0.025$. Compare the results with those found in parts (a) and (b).
- (d) Find the exact solution $y = \phi(t)$ of the initial value problem and evaluate $\phi(t)$ at $t = 0.1, 0.2, 0.3, 0.4$. Compare these values with the results of parts (a), (b), and (c).

Solution. (a)–(c) For Euler's method,

$$t_n = nh, \quad y_{n+1} = y_n + h(1 - t_n + 2y_n), \quad y_0 = 1.$$

Carrying out the recurrence gives

h	$y(0.1)$	$y(0.2)$	$y(0.3)$	$y(0.4)$
0.100	1.30000	1.65000	2.06000	2.54200
0.050	1.31250	1.68013	2.11445	2.62949
0.025	1.31938	1.69682	2.14482	2.67859

Thus, for each listed value of t , the Euler approximations increase as the step size decreases.

(d) The differential equation can be written as

$$y' - 2y = 1 - t.$$

An integrating factor is e^{-2t} , so

$$(e^{-2t}y)' = (1 - t)e^{-2t}.$$

Hence

$$y = ce^{2t} + \frac{t}{2} - \frac{1}{4}.$$

Using $y(0) = 1$ gives $c = 5/4$, and therefore

$$\boxed{\phi(t) = \frac{5}{4}e^{2t} + \frac{t}{2} - \frac{1}{4}}.$$

The corresponding exact values are

	$\phi(0.1)$	$\phi(0.2)$	$\phi(0.3)$	$\phi(0.4)$
Exact	1.32675	1.71478	2.17765	2.73193

The Euler approximations move toward these exact values as h decreases. □

Problem 2.8.2. Transform the initial value problem

$$\frac{dy}{dt} = 4 - y^3, \quad y(-1) = 2,$$

into an equivalent problem with the initial point at the origin.

Solution. Set

$$s = t + 1, \quad w = y - 2.$$

Then the initial point $(t, y) = (-1, 2)$ becomes $(s, w) = (0, 0)$, and

$$\frac{dw}{ds} = \frac{dy}{dt}.$$

Since $y = w + 2$, the equivalent initial value problem is

$$\boxed{\frac{dw}{ds} = 4 - (w + 2)^3, \quad w(0) = 0.}$$

□

Problem 2.8.7. Let $\phi_0(t) = 0$ and use the method of successive approximations to approximate the solution of

$$y' = 2t^2 + y^2, \quad y(0) = 0.$$

- (a) Calculate $\phi_1(t), \phi_2(t), \phi_3(t)$.
- (b) Plot $\phi_1(t), \phi_2(t), \phi_3(t)$. Observe whether the iterates appear to be converging.

Solution. (a) The successive approximations satisfy

$$\phi_{n+1}(t) = \int_0^t (2s^2 + \phi_n(s)^2) \, ds.$$

Thus

$$\begin{aligned} \phi_1(t) &= \int_0^t 2s^2 \, ds = \frac{2}{3}t^3, \\ \phi_2(t) &= \int_0^t \left(2s^2 + \frac{4}{9}s^6 \right) \, ds = \frac{2}{3}t^3 + \frac{4}{63}t^7, \end{aligned}$$

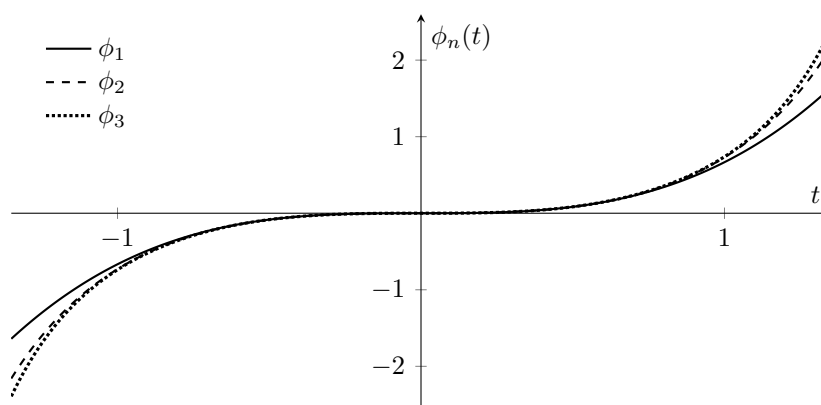
and

$$\begin{aligned} \phi_3(t) &= \int_0^t \left(2s^2 + \left(\frac{2}{3}s^3 + \frac{4}{63}s^7 \right)^2 \right) \, ds \\ &= \frac{2}{3}t^3 + \frac{4}{63}t^7 + \frac{16}{2079}t^{11} + \frac{16}{59535}t^{15}. \end{aligned}$$

Hence

$$\boxed{\begin{aligned} \phi_1(t) &= \frac{2}{3}t^3, \\ \phi_2(t) &= \frac{2}{3}t^3 + \frac{4}{63}t^7, \\ \phi_3(t) &= \frac{2}{3}t^3 + \frac{4}{63}t^7 + \frac{16}{2079}t^{11} + \frac{16}{59535}t^{15}. \end{aligned}}$$

(b) The first three iterates are shown below.



Near the initial point, the successive iterates lie increasingly close to one another, which is consistent with convergence of the Picard iterates there.

□

Problem 3.1.3. Find the general solution of

$$12y'' - y' - y = 0.$$

Solution. The characteristic equation is

$$12r^2 - r - 1 = (3r - 1)(4r + 1) = 0,$$

so

$$r_1 = \frac{1}{3}, \quad r_2 = -\frac{1}{4}.$$

Therefore

$$y = c_1 e^{t/3} + c_2 e^{-t/4}.$$

□

Problem 3.1.4. Find the general solution of

$$y'' + 6y' = 0.$$

Solution. The characteristic equation is

$$r^2 + 6r = r(r + 6) = 0,$$

with roots $r_1 = 0$ and $r_2 = -6$. Hence

$$y = c_1 + c_2 e^{-6t}.$$

□

Problem 3.1.9. Find the solution of the initial value problem

$$y'' + 3y' = 0, \quad y(0) = 0, \quad y'(0) = 3.$$

Sketch the graph of the solution and describe its behavior as t increases.

Solution. The characteristic equation is

$$r(r + 3) = 0,$$

so the general solution is

$$y = c_1 + c_2 e^{-3t}.$$

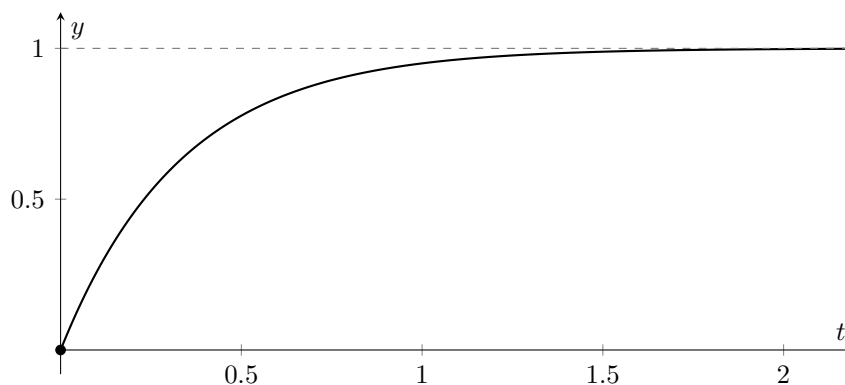
The initial conditions give

$$c_1 + c_2 = 0, \quad -3c_2 = 3,$$

so $c_1 = 1$ and $c_2 = -1$. Therefore

$$\boxed{y = 1 - e^{-3t}}.$$

The graph for $t \geq 0$ is shown below; the dashed line is the horizontal asymptote $y = 1$.



Since $e^{-3t} \rightarrow 0$ as $t \rightarrow \infty$,

$$\boxed{y(t) \rightarrow 1 \text{ as } t \rightarrow \infty}.$$

The solution increases from 0 and approaches 1 from below.

□

Problem 3.2.7. Determine the longest interval in which the initial value problem

$$t(t - 4)y'' + 3ty' + 5y = 2, \quad y(3) = 0, \quad y'(3) = -1$$

is certain to have a unique twice-differentiable solution. Do not attempt to find the solution.

Solution. Dividing by $t(t - 4)$ gives

$$y'' + \frac{3}{t-4}y' + \frac{5}{t(t-4)}y = \frac{2}{t(t-4)}.$$

The coefficients are continuous except at $t = 0$ and $t = 4$. The largest open interval containing the initial point $t = 3$ on which all coefficients are continuous is

$$\boxed{(0, 4)}.$$

By Theorem 3.2.1, the initial value problem has a unique twice-differentiable solution throughout this interval.

□

Problem 3.2.18. Consider the differential equation

$$y'' - y' - 2y = 0.$$

(a) Show that $y_1(t) = e^{-t}$ and $y_2(t) = e^{2t}$ form a fundamental set of solutions.

(b) Let

$$y_3(t) = -2e^{2t}, \quad y_4(t) = y_1(t) + 2y_2(t), \quad y_5(t) = 2y_1(t) - 2y_3(t).$$

Are $y_3(t)$, $y_4(t)$, and $y_5(t)$ also solutions of the differential equation?

(c) Determine whether each of the pairs $\{y_1(t), y_3(t)\}$ and $\{y_2(t), y_3(t)\}$ forms a fundamental set of solutions.

Solution. (a) Direct substitution gives

$$y_1'' - y_1' - 2y_1 = e^{-t} + e^{-t} - 2e^{-t} = 0$$

and

$$y_2'' - y_2' - 2y_2 = 4e^{2t} - 2e^{2t} - 2e^{2t} = 0,$$

so both functions are solutions. Their Wronskian is

$$\begin{aligned} W(y_1, y_2)(t) &= \begin{vmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{vmatrix} \\ &= 3e^t \neq 0. \end{aligned}$$

Hence y_1 and y_2 form a fundamental set of solutions.

(b) Since $y_3 = -2y_2$, $y_4 = y_1 + 2y_2$, and

$$y_5 = 2y_1 - 2y_3 = 2y_1 + 4y_2,$$

each is a linear combination of the solutions y_1 and y_2 . By Theorem 3.2.2, all three are also solutions.

(c) Since $y_3 = -2y_2$,

$$W(y_1, y_3)(t) = -2W(y_1, y_2)(t) = -6e^t \neq 0,$$

so

$$\boxed{\{y_1, y_3\} \text{ is a fundamental set}}.$$

On the other hand, y_2 and y_3 are constant multiples of one another, so their Wronskian is zero. Thus

$$\boxed{\{y_2, y_3\} \text{ is not a fundamental set}}.$$

□

Problem 3.2.20. Find the fundamental set of solutions specified by Theorem 3.2.5 for

$$y'' + 5y' + 4y = 0, \quad t_0 = 1.$$

Solution. The characteristic equation is

$$r^2 + 5r + 4 = (r + 1)(r + 4) = 0,$$

so it is convenient to write the general solution as

$$y = Ae^{-(t-1)} + Be^{-4(t-1)}.$$

By Theorem 3.2.5, y_1 is the solution satisfying

$$y_1(1) = 1, \quad y_1'(1) = 0.$$

Thus

$$A + B = 1, \quad -A - 4B = 0,$$

which gives $A = 4/3$ and $B = -1/3$. Hence

$$y_1(t) = -\frac{1}{3}e^{-4(t-1)} + \frac{4}{3}e^{-(t-1)}.$$

Similarly, y_2 is the solution satisfying

$$y_2(1) = 0, \quad y_2'(1) = 1.$$

Then

$$A + B = 0, \quad -A - 4B = 1,$$

so $A = 1/3$ and $B = -1/3$. Therefore the specified fundamental set is

$$\begin{aligned} y_1(t) &= -\frac{1}{3}e^{-4(t-1)} + \frac{4}{3}e^{-(t-1)}, \\ y_2(t) &= -\frac{1}{3}e^{-4(t-1)} + \frac{1}{3}e^{-(t-1)}. \end{aligned}$$

□

Problem 3.3.6. Find the general solution of

$$y'' - 2y' + 8y = 0.$$

Solution. The characteristic equation is

$$r^2 - 2r + 8 = 0,$$

with roots

$$r = 1 \pm i\sqrt{7}.$$

Therefore

$$y = c_1 e^t \cos(\sqrt{7}t) + c_2 e^t \sin(\sqrt{7}t).$$

□

Problem 3.3.13. Find the solution of the initial value problem

$$y'' - 2y' + 5y = 0, \quad y\left(\frac{\pi}{2}\right) = 0, \quad y'\left(\frac{\pi}{2}\right) = 4.$$

Sketch the graph of the solution and describe its behavior for increasing t .

Solution. The characteristic equation is

$$r^2 - 2r + 5 = 0,$$

with roots $r = 1 \pm 2i$. Hence

$$y = e^t(c_1 \cos 2t + c_2 \sin 2t).$$

The first initial condition gives $c_1 = 0$. Thus

$$y = c_2 e^t \sin 2t,$$

and

$$y' = c_2 e^t (\sin 2t + 2 \cos 2t).$$

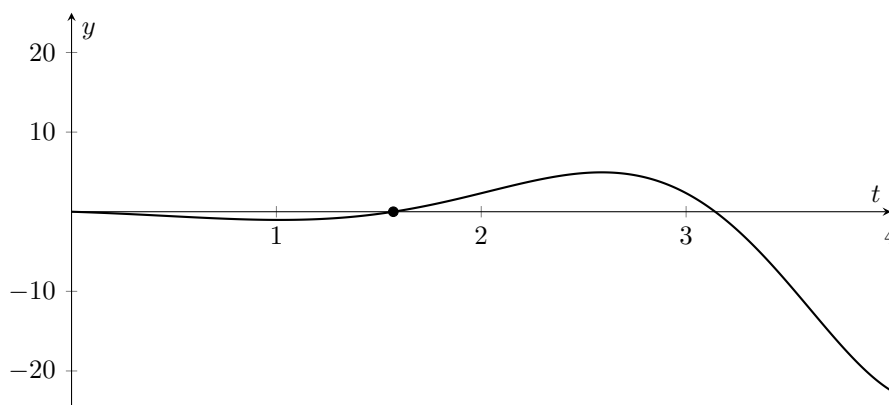
Using $y'(\pi/2) = 4$ gives

$$-2c_2 e^{\pi/2} = 4, \quad c_2 = -2e^{-\pi/2}.$$

Therefore

$$y = -2e^{t-\pi/2} \sin 2t.$$

The graph is shown below. The marked point is the initial point $(\pi/2, 0)$.



The trigonometric factor produces oscillation with period π , while the factor $e^{t-\pi/2}$ makes the amplitude grow exponentially. Thus the solution is a

growing oscillation.

□

Problem 3.4.4. Find the general solution of

$$4y'' - 12y' + 9y = 0.$$

Solution. The characteristic equation is

$$4r^2 - 12r + 9 = (2r - 3)^2 = 0,$$

so $r = 3/2$ is a repeated root. Therefore

$$y = c_1 e^{3t/2} + c_2 t e^{3t/2}.$$

□

Problem 3.4.19. Use the method of reduction of order to find a second solution of

$$t^2 y'' + t y' - 4y = 0, \quad t > 0,$$

given that $y_1(t) = t^2$ is one solution.

Solution. Set

$$y = v(t)y_1(t) = v(t)t^2.$$

Then

$$y' = v't^2 + 2vt, \quad y'' = v''t^2 + 4v't + 2v.$$

Substitution into the differential equation gives

$$t^4v'' + 5t^3v' = 0.$$

Since $t > 0$, this reduces to

$$tv'' + 5v' = 0.$$

Let $w = v'$. Then

$$tw' + 5w = 0,$$

so

$$\frac{w'}{w} = -\frac{5}{t}.$$

Hence

$$w = ct^{-5}, \quad v = -\frac{c}{4}t^{-4} + k.$$

Therefore

$$y = vt^2 = -\frac{c}{4}t^{-2} + kt^2.$$

The term kt^2 is a multiple of the known solution y_1 , while t^{-2} is not a constant multiple of t^2 . Thus an independent second solution is

$$\boxed{y_2(t) = t^{-2}}.$$

□